

Effect of Electroconvection and Its Use in Intensifying the Mass Transfer in Electrodialysis (Review)¹

V. V. Nikonenko^{a, *}, S. A. Mareev^a, N. D. Pis'menskaya^a, A. M. Uzdenova^b, A. V. Kovalenko^a,
M. Kh. Urtenov^a, and G. Pourcelly^c

^aKuban State University, Krasnodar, 350040 Russia

^bKarachaevsk-Cherkessky State University Named after U.D. Aliev, Karachaevsk, 369202 Russia

^cInstitute European of the Membranes, University of Montpellier, ENSCM, CNRS, CC047, Montpellier, 34095 France

*e-mail: nikon@chem.kubsu.ru

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Abstract—The modern concepts on the origin of electroconvection (EC) are surveyed briefly and the known mechanisms of this phenomenon are classified. Factors that influence the EC character and intensity at the surface of ion-exchange membranes are analyzed, such as electrical and geometrical heterogeneity of the membrane surface, its degree of hydrophobicity, and the surface charge. The EC mechanism is shown also to depend on the applied potential difference and the rate of solution flow between membranes. The mechanism of the EC-induced gain in the mass transfer is elucidated, the possible gain in the mass transfer is estimated, and the prospects for using the EC for reducing the membrane fouling caused by sedimentation and formation of organic deposits are assessed.

Keywords: electroconvection, overlimiting current mode, ion-exchange membranes, mathematical simulations

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INTRODUCTION

The separation processes in which the electric current provides the transport of not only ions but also water find ever increasing number of new applications. These processes are based on electroconvection (EC) where, under the action of the electric force, the liquid molecules are carried along with the ions that form the

space charge at the ion-selective surface. The intensity of EC can be considerably increased by passing the overlimiting current when, due to polarization of the electric double layer (EDL), the extended macroscopic space charge region is formed on the interface. The studies in this field date back to 1949 when Levich turned his attention to polarization of the EDL at the electrode surface when electric current is passed.

The intense current modes both stationary and nonstationary appear to be a promising direction for the development of electrodialysis (ED) [1, 2]. Their use makes it possible to substantially increase the mass-transfer rate and, hence, decrease the surface of expensive ion-exchange membranes (IEM), which may provide noticeable economy [3]. Moreover, the use of pulse current allows to substantially decrease the membrane fouling, i.e., sedimentation of poorly soluble salts and/or formation of mineral and organic/colloidal deposits on the membrane surface and in its pores [4–6]. The intense current modes are also used in microfluid devices [7], in capacitive deionization [8], in microanalytical chemistry [9]. Microfluid devices operate with small amounts of liquids. They are used in separation of macromolecules, including DNA, separation of disperse particles according to their size and conductivity, for development of new materials, formation of images on screens of iPhones,

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electronic books, and development of micropumps and micromixers, etc. [10]. The electrodialysis microdevices allow solutions of proteins and peptides to be concentrated by a factor of 1 million [11], which attracts attention of analytic chemists.

The principle of functioning of a classical ED is based on the transfer of ions under the action of the external electric field. In electrokinetic devices, the main result of electric-field action is the transfer of a solution volume. A characteristic feature of ED and other devices listed above when they operate in intense current modes is that the electric field induces the motion of both the ions and the solution volume. The latter factor is very important because provides a new approach to the design and working mode of electromembrane systems. In the recent five years, the peculiarities of operation of systems such as the ion-concentration polarization ED (ICP ED) in which only one IEM type is used [12], the shock ED [13–16] which employs porous materials at the interface with IEM [17], and other electrodialysis microdevices [18] were being proposed and actively discussed. The transport phenomena involving electroconvection in nano-microfluid systems were thoroughly surveyed in several reviews [19–21]. Stone et al. [19] surveyed the hydrodynamic aspects of functioning of such devices which lie in the basis of “lab-on-chip” devices by taking into account the geometrical and chemical characteristics of the surface. Schoch et al. [20] reviewed the physics of transport phenomena in nanofluid systems with if only one characteristic size <100 nm in which the phenomena impossible in the larger-scale systems may occur. Chang et al. [21] presented a review of application of microfluidics to biology, medicine, and analytic chemistry with the use of nano-microchannel devices and ion-exchange membranes and also analyzed the effect of electrokinetic microvortices on the ion transport in such devices [21–23]. Bazant et al. [24] considered the specific features of electrokinetic phenomena at high voltage [24].

For electromembrane systems used in separation processes (electrodialysis, electrodeionization), the factor of interest is the possibility of enhancing the ion transfer rate, especially in dilute solutions in which the limiting current density (i_{lim}) is low [1, 2]. Recently, owing to attempts of many scientists, it was found that electroconvection is the main mechanism of the overlimiting current through IEM in ED when dilute solutions are processed [25–40]. In the general case, EC represents a phenomenon of liquid transport under the action of electric force applied to the space charge of a depleted solution near the ion-selective surface [41–43].

Frilette [25] discovered that in membrane systems, the current density may exceed its limiting value; moreover, this surplus is associated with the increase in mass transport through the membrane and can be ascribed to neither the loss of selectivity nor the

appearance of new carriers (H^+ and OH^- ions) as a result of water splitting. Frilette explained this increase by the current-induced convection, which he called the “electrogravitational transport.” Dukhin et al. [30] observed the unlimited growth of current through an ion-exchange granule upon the increase in voltage, which was explained by electroosmosis of the second kind [31]. By changing the position of the cell with a ion-exchange membrane in Earth’s gravitational field, Rubinstein et al. [44] separated the contribution of gravitational convection into the overlimiting transport. Malezki et al. [33] demonstrated that deposition of a layer of uncharged agar–agar gel on the ion-exchange membrane surface suppresses the overlimiting transport of ions in a nonflow cell (a modification of the cell used in [44]). Insofar as the mechanism of ion transport in a gel layer differs from that in solution only by the absence of convection, the authors [33] arrived at a conclusion that the overlimiting transport is associated with the natural convection induced by current. Rubinstein et al. [45] carried out the analogous, albeit more precise, studies which made it possible to conclude that electroconvection is the origin of the natural convection inducing the overlimiting transfer in membrane systems described in [33, 44, 45]. Zabolotsky et al. [35, 36] determined the conditions of the ED process in which the gravitational convection or electroconvection dominated. In concentrated solutions, the gravitational convection may play a significant role [35, 37, 46–49].

In this brief review we present the modern views on the EC phenomenon, its main mechanisms, and the factors that may affect its occurrence in membrane systems. These factors include the current density/potential difference, the hydrodynamic conditions and also the structure and properties of the IEM surface. We will discuss how one can control the EC and the prospects of using this phenomenon for ED modernization.

MECHANISMS OF ELECTROCONVECTION

Although EC in membrane systems is of the same origin as the electroosmosis (EO) (discovered in 1807 by Reuss [42]), there is substantial difference between these two phenomena. A quantitatively significant electroosmotic liquid flow takes place in a capillary under tangentially applied electric field; the rate of EO flow is proportional to the applied field strength. The EC appears when current is passed in the normal direction to the membrane surface; moreover, the distance between the neighboring membranes is of no substantial importance. The important factor for understanding EC is that the thickness and the structure of the electric double layer at the membrane(electrode)/solution interface depend on the polarizing electric current flowing perpendicularly to the interface. Levich was the first to notice the latter fact. Back in the 40s–50s of the XX century he showed [50, 51]

that small currents passed in an electroneutral solution shift the electrolyte concentration at the electrode surface: however, this region retains the Boltzmann distribution of counterions and coions. Levich's ideas were developed further by Grafov, Chernenko, Dukhin, Shilov, Newman, Smyrl, Rubinstein, and others [25, 42, 52–57]. It deserves mention that the key work in this series is that by Rubinstein and Shtilman [26] in which they solved the equations of Nernst–Planck and Poisson without using the condition of local electroneutrality. Their solution showed that with the high voltage applied, the ion concentration at the ion-selective surface does not tend to zero (as observed when the electroneutrality condition is fulfilled). Instead, the space-charge region (SCR) increases due to the appearance of a nonequilibrium zone in which the Boltzmann distribution of concentration is not fulfilled. The extended SCR can reach several micrometers [58] which is wider than the EDL thickness by two–three orders of magnitude. The concentration of counterions in the extended SCR changes insignificantly (Fig. 1) being of the order of $c_0 (\lambda/\delta)$, where c_0 is the concentration of counterions in solution bulk, λ is the EDL thickness, and δ is the diffusion layer thickness [25, 58–60]. The relatively high space charge values (determined by the counterion concentration) and the extended-SCR thickness make possible the appearance of very intense EC [33, 61, 62]. Note that the charge of the extended SCR is almost independent of the charge of the ion-selective surface.

The mathematical modeling of the transport of ions and water with account for EC near the ion-selective surface is based on the system of equations of Nernst–Planck–Poisson–Navier–Stokes (NPP–NS) [28, 36, 39, 57, 63–72]. The models based on this system of equations are two-dimensional and, in certain cases [73, 74], three-dimensional.

Figure 2 compares the experimental voltammetric characteristics (VAC) for homogeneous Neosepta CMX (Japan) and heterogeneous MK-40 (Russia) cation-exchange membranes. Two systems of equations were used: NPP–NS and the equations of Nernst–Planck–Navier–Stokes (NP–NS). In the latter case, the Poisson equation is replaced by the condition of local electroneutrality. This model was put forward by Sonin and Probstein [75] and further developed by Gnusin et al. [76] and Shaposhnik et al. [77]. The solutions obtained with the use of both systems of equations have the almost identical VAC in the region of underlimiting currents (Fig. 2). If the membrane surface is homogeneous and its charge is not too high, then the contribution of EC at $i < i_{lim}$ into the ion transport and, hence, into the VAC shape is insignificant. In this case, the curve is linear. However, it deserves mention that the name “ohmic region” which is often used in the literature for this VAC part is not quite correct, strictly speaking. Ohm's law

assumes not only the linear dependence of current on voltage but also the instantaneous response of the current to the voltage changes (or instantaneous response of the voltage to the changes in current). In fact, the potential drop in the membrane system contains, besides the ohmic component, the potential drop of the diffusion nature proportional to logarithm of the ratio of the concentration near the membrane surface c_s to the concentration in solution bulk c_0 [78, 79]. It also includes the diffusion potential drop in diffusion layers adjoining the membrane and the Donnan potential drops at both interfaces. Upon sharp changes in the current, the concentration cannot change instantaneously; hence, the response of the corresponding voltage components delays, whereas the ohmic component reacts immediately. For low currents ($i \ll i_{lim}$), under (quasi)stationary conditions, $\ln(c_s/c_0)$ can be expressed by the ratio i/i_{lim} [78], which explains the linear form of the dependence of the sum of diffusion potentials on the current.

When the potential difference across the IEM approaches 0.5 V, the current density, according to the NP–NS model, approaches its limiting value i_{lim} , whereas the experimental VAC and also the VAC calculated in terms of the NPP–NS model demonstrate no saturation. According to the model, the gain in the current is due to the EC development. The possible mechanisms of the latter are shown below.

After reaching the limiting current, an inclined plateau appears in the VAC, called in the literature the “limiting-current plateau” [25, 80]. The intersection of the tangents drawn to the initial linear VAC region and to the inclined plateau provides the highly accurate value of i_{lim} identical in terms of both models, i.e., NPP–NS and NP–NS. As the current/voltage increases, oscillations appear in the VAC plateau, after which the current increases sharply at a relatively small increase in the voltage (Fig. 2).

The VAC of an ion-exchange membrane strongly depends on its surface properties. Figure 2 shows that the curves of homogeneous CMX and heterogeneous MK-40 membranes differ substantially. Hence, when comparing calculated and experimental curves, one can expect only qualitative agreement. The model proposed by Urtenov et al. [69] and based on the NPP–NS equations is called the “basic” because uses only fundamental laws and contains neither empiric formulas nor adjustable parameters. It can be stated that this model describes an “ideal” system with a homogeneous ideally-selective membrane. Such models are called “first principles models.” To attain the quantitative agreement with experimental VAC, one has to take into account the concrete features of the surface of a given membrane (its surface geometry, the distribution of conducting and nonconducting regions, hydrophilic and hydrophobic regions, etc.).

At present, we know that the EC can develop according to different mechanisms depending on the

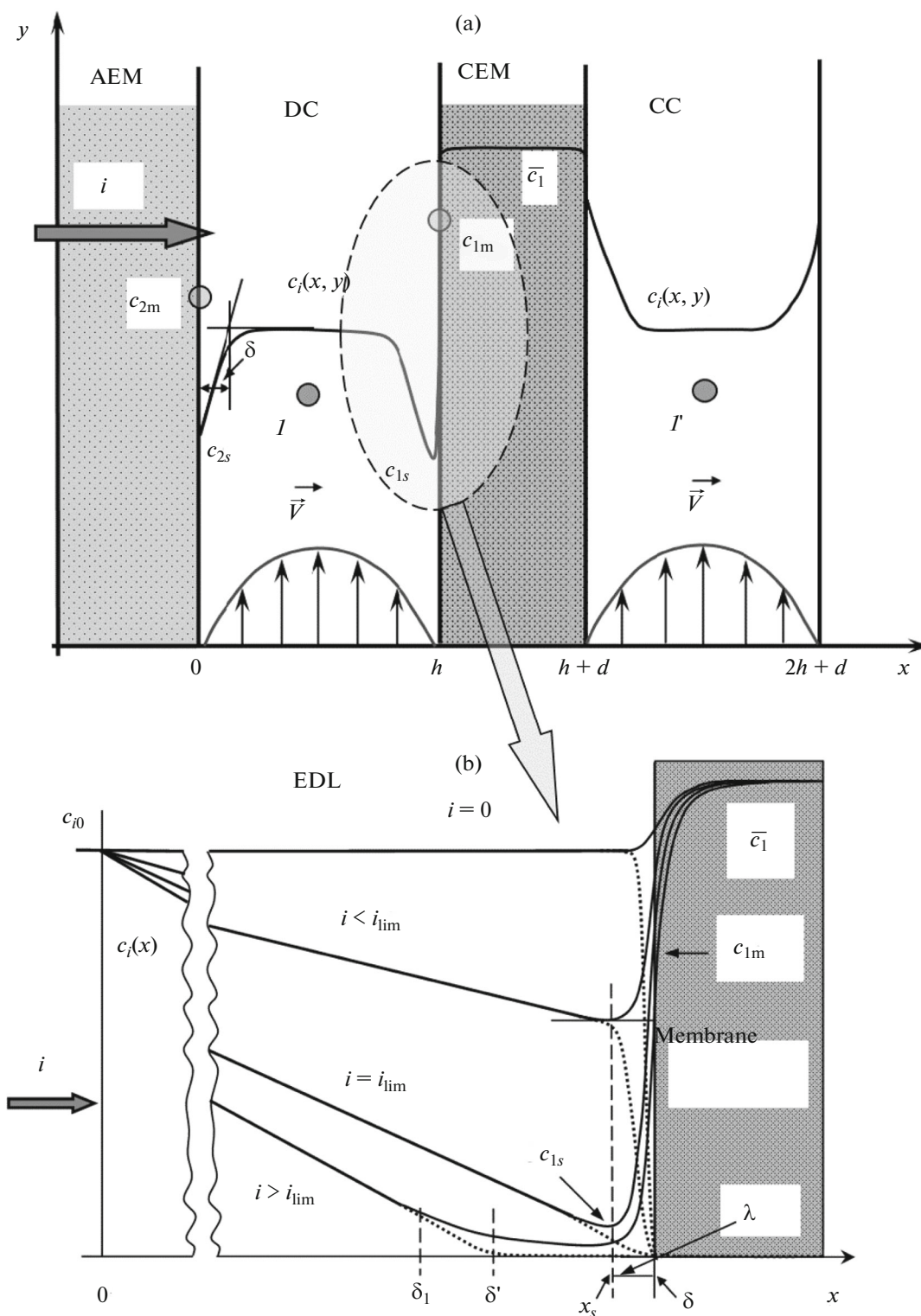


Fig. 1. (a) Illustration of an electrodialysis chamber pair which consists of concentration (CC) and desalination (DC) chambers divided by an anion-exchange (AEM) and cation-exchange (CEM) membranes. Points I and I' indicate the position of Luggin capillaries used for detecting the potential difference on the membrane under study. Concentration profiles are shown schematically; c_{1s} is the minimum cation concentration near the CEM, c_{1m} and $\bar{c}_1$ are the cation concentrations in the sliding plane and in the CEM volume, respectively. (b) Illustration of the concentration profiles of counterions (solid lines) and coions (dashed lines) in the diffusion layer with the thickness δ near the CEM at different current densities. The diffusion layer is divided into two regions: electroneutral ($0 \leq x \leq \delta_1$) and SCR ($\delta_1 \leq x \leq \delta$). The extended part of SCR is between points δ_1 and x_s , the quasi-equilibrium part of SCR is between x_s and δ . The intersection of the line which extends the concentration profile linear part with the axis x gives the effective thickness of the diffusion layer (δ').

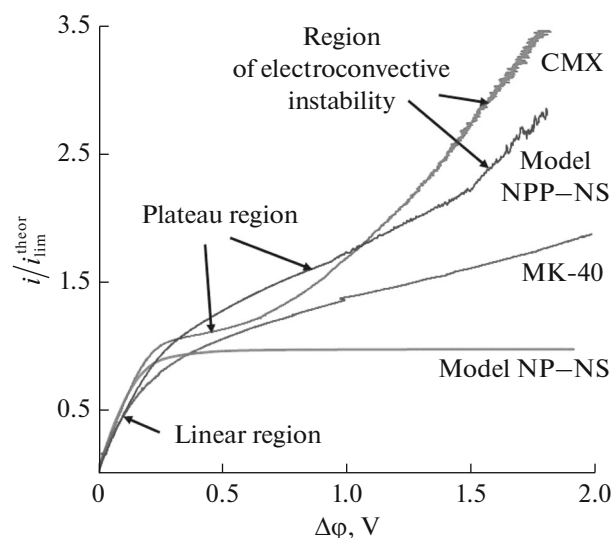


Fig. 2. Experimental and calculated voltammetric characteristics of ion-exchange membranes. Membrane brands are shown near the curves. Calculations were carried out using equations of Nernst–Planck–Poisson–Navier–Stokes (NPP–NS) and equations of Nernst–Planck–Navier–Stokes (NP–NS). In the second system, the Poisson equation is changed for the local electroneutrality condition. Adapted from [69].

applied voltage and other conditions (Fig. 3). Theoretically [28, 45, 81] one can distinguish two main mechanisms: (1) the bulk EC caused by electric field action on the residual space charge in a quasi-electroneutral electrolyte solution with nonuniform distribution of concentration and (2) the EC caused by electroosmotic sliding of the SCR and formed in the depleted solution at the interface with an ion-exchange surface of electrode or membrane. The EC developed by both mechanisms can be of two kinds. At the low current/voltage when the EDL on the interface is (quasi)equilibrium (i.e., the structure of its diffuse part retains the Boltzmann distribution of ions shifted under the effect of the external field [51]), the (quasi)equilibrium EC occurs. In the case of electroosmotic mechanism, this kind of EC is also called the EC of the first kind (according to terminology by Dukhin and Mishchuk [25, 33]). In contrast to the equilibrium EC [81], in overlimiting modes the non-equilibrium EC is developed. In this case, the EO of the first kind transforms into EO of the second kind [64, 82]. This type of EC is characterized by the presence of an extended SCR localized beyond the quasi-equilibrium EDL (Fig. 1). In the extended SCR, the counterion concentration passes through a minimum c_{1s} as we approach the interface after which quickly increases approaching the value in the membrane volume close to the concentration of fixed ions. Near the interface, the SCR structure does not differ qualitatively from that of the equilibrium EDL.

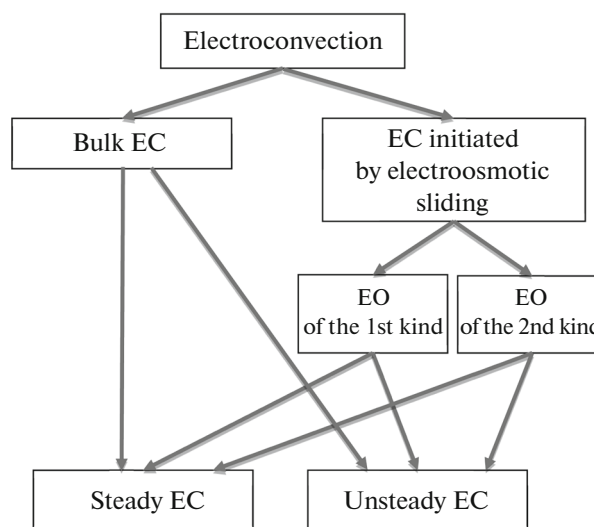


Fig. 3. Classification of EC mechanisms.

The development of EO of both the first and the second kind requires the presence of a tangential electric field the action of which on the SCR gives rise to EO sliding. The tangential field can be formed due to electrical heterogeneity and/or curvature of the surface [25, 33, 45] as well as due to nonuniformity of the concentration field distribution [69]. Furthermore, note that the EO flow can appear as a result of spontaneous perturbations of the originally homogeneous electric field [28]. For the forced solution flow in the electrodialyzer channels, the nonuniformity of the concentration field can be associated with solution desalination which increases as the solution moves ahead in the channel. This increases the solution resistance along the transverse coordinate and, as a consequence, the current lines concentrate near the channel inlet [69].

The EO of the first and second kinds can appear in both the thresholdless and threshold modes. In the first case, we observe steady EC due to the action of a stable tangential field on the SCR the main part of which represents the quasi-equilibrium EDL. The other EC type develops in the membrane system at relatively low current/voltage (corresponding to the linear part of VAC and a part of the inclined plateau where the curve is smooth, Fig. 2). Usually the contribution of steady EC into the mass transition is insignificant. In the second case, the hydrodynamically unsteady EC is developed. It was I. Rubinstein and B. Zaltzman [28] who were the first to propose the mechanism of EC unsteadiness. The unsteadiness observed upon reaching a certain threshold voltage value is caused by the appearance of positive feedback between the fluctuation of the local tangential force and the EO-sliding rate [28, 45, 73]. The tangential force is directed from the depleted solution region at the membrane surface to the region in which solution

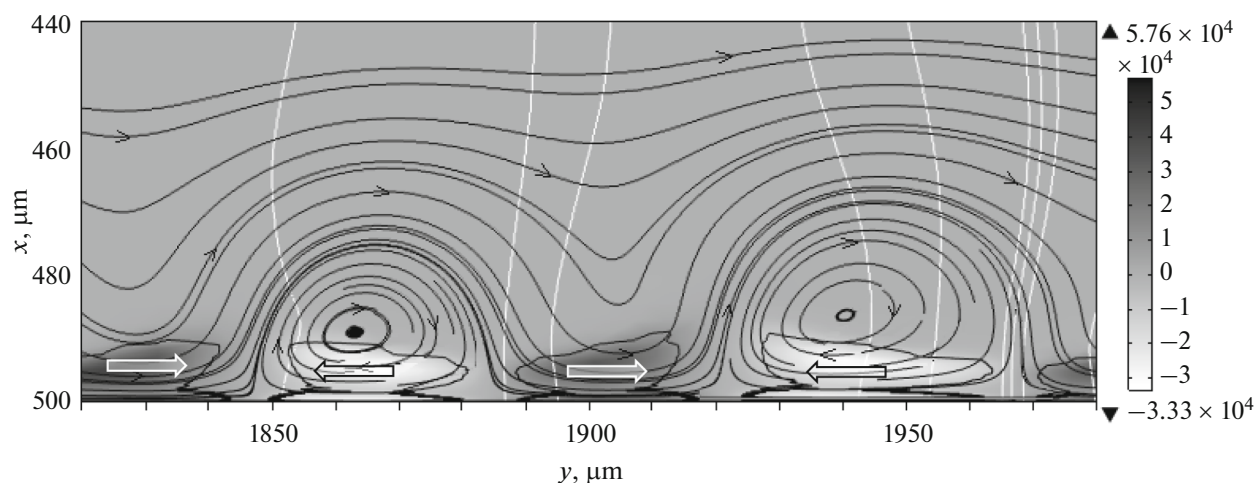


Fig. 4. Distribution of the tangential component of electric force (its magnitude is shown by shades of grey and its direction is indicated by arrows) and also of solution flow lines (black lines) and current flow lines (white lines) in the overlimiting state. Adapted from [83].

is enriched due to rotation of a vortex which brings the relatively concentrated solution from the bulk (Fig. 4). If the force increases for some reason, the vortex rotation velocity increases together with the electrolyte concentration at the surface, which, in turn, increases in the tangential force (Fig. 4).

The unsteady EC induces the current/voltage oscillations. In VAC shown in Fig. 2, the oscillations in both experimental and calculated (NPP–NS model) curves start at a voltage of about 1 V. However, in the other cases (high charge and surface heterogeneity, multicharged counterions) described below, such oscillations may start at the normalized potential drop of 30–50 mV. In this case, an anomalous region appears in the stationary VAC in this potential-drop region, where the potential does not increase as usual but decreases with the increase in current.

E. Zholkovskij et al. [84] theoretically showed that near an ideally selective smooth homogeneous membrane, the EC arising by the mechanism of EO of the first kind under usual conditions can proceed only in the steady mode. The electro-hydrodynamic instability may occur only when large ions are present in solution (with diameter >40 nm). However, if the membrane is not ideally selective, the EO of the first kind may lead to hydrodynamic instability, as was shown recently by I. Rubinstein and B. Zaltzman [81, 85]. The reason for this is the gradient of electric potential in the tangential direction which appears in this case.

The theoretical studies of bulk EC are sufficiently numerous [57, 86, 87], including the studies by A. Grigin and A. Davydov [57, 87]. In the literature, the discussion of whether the instability of this kind of EC is possible was sufficiently long. I. Lerman et al. [88] showed that for an ideally selective membrane, the bulk electroconvective instability cannot arise. However, if the membrane is not ideally selective,

then, as was shown by I. Rubinstein et al. [81, 85], the electric field impact on the residual space charge in a quasi-electroneutral solution may lead to electroconvective instability.

According to the theory [81, 85, 88], when EC originates by the electroosmotic mechanism, the sliding of liquid in SCR along the membrane surface is the necessary condition. At the same time, the bulk EC can also occur if the liquid in the SCR is immobile. Thus, if we immobilize solution in SCR, then the EC can arise only by the mechanism of bulk electroconvection. An experiment of such kind was described by I. Rubinstein et al. [45]. The surface of a cation-exchange membrane was covered with a layer of polyvinyl alcohol (PVA) chemically bound with the surface. Such a layer, like an agar–agar layer [33], can immobilize the solution by making it convectively immobile. However, it carries no fixed charges; hence, the electrodiffusion transfer of ions in it proceeds in the same manner as in solution. The VAC were obtained for a series of membranes with different thickness of the immobilizing layer. As the layer thickness increased, the overlimiting transfer through the membrane decreased disappearing at a thickness of 1–2 μm . It is evident that electroconvection disappears when all space charge is immobilized inside the PVA layer, i.e., the electroosmotic sliding disappears. Such a behavior bears witness of the electroosmotic mechanism of overlimiting transfer under experimental conditions [45], in contrast to the mechanism of bulk electroconvection [45]. Note that according to other estimates [58], the SCR thickness at the membrane surface at voltages used in the experiment [45] reached 1–2 μm , which adequately agreed with the limiting thickness of immobilized layer. The electroosmotic mechanism was also confirmed by the fact that the form of VAC and the behavior of IEM in the

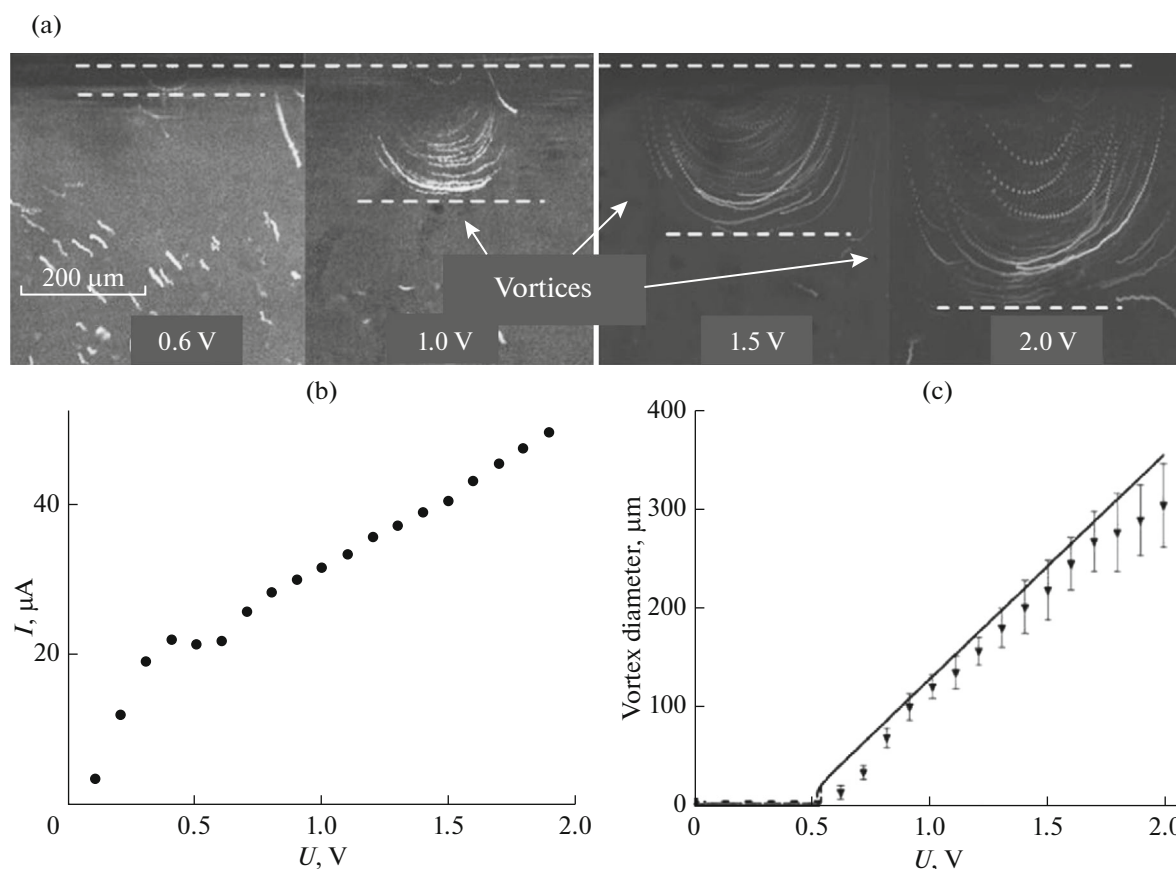


Fig. 5. (a) Image of solution in a cell with transparent walls. The membrane is at the upper boundary of the image, the trajectories of liquid flow are visualized by means of tracer particles; the voltage values (in V) are shown near the corresponding vortices. (b) Voltammetric characteristic. (c) The size of the vortex structure depending on the applied voltage (adapted from [89]).

region of overlimiting (and sometimes underlimiting) currents depend strongly on its surface properties (which should not be the case of bulk electroconvection, see the Section devoted to the effect of surface properties of IEM).

EXPERIMENTAL STUDIES OF ELECTROCONVECTION

The studies of the effect of forced convection and gravitational convection on the mass transfer and the diffusion-layer structure at the ion-selective surface have a long history, numerous publications were devoted to this subject [15, 16, 25]. Publications that provide the experimental evidence of electroconvection phenomenon, especially, its visualization are much fewer. For the first time, the EC visualization was described by S. Kim et al. [7]. These authors studied the steady EC mode associated with EO flows in a micro/nanofluid system in solution near a nanoporous membrane. The voltage region corresponding the thresholdless transition from quasi-equilibrium osmotic flow to nonequilibrium flow was studied. To visualize liquid flows, they used tracer par-

ticles. The first attempt to visualize the EC in the unsteady mode was undertaken by S. Rubinstein et al. [89]. Their study was carried out in a copper sulfate solution at a surface of a cation-exchange membrane with the use of a nonflow cell. The current was supplied by means of a copper anode separated from membrane by solution and a copper cathode placed on the other side of the membrane. To avoid the gravitational convection, the membrane was placed horizontally in such a way that the depleted less dense solution found itself immediately under the membrane. As in [7], the liquid flows were visualized by means of tracer particles with simultaneous VAC measurements (Fig. 5). According to Fig. 5, the noticeable vortices appear immediately after reaching the limiting state corresponding to the appearance of the plateau in the VAC.

G. Yossifon and H. Chang [90] visualized the vortex structures in a nano-microfluid system with applied alternating current. R. Kwak et al. [91] carried out visualization of EC vortices in a microelectrodialysis cell with forced flow. I. Green et al. [92] studied visually the effect of the heterogeneous micro-nano-structure on the appearance of vortices. I. De Valença

et al. [93] studied dynamics of the development of vortices near an IEM in a nonflow cell similar to that used by S. Rubinstein [89].

V. Shaposhnik, V. Vasil'eva et al. [94–97] visualized concentration profiles in an ED cell with forced flow by means of laser interferometry. They showed [94] that the overall thickness of the diffusion layer (determined as the distance from the membrane to the point in which the salt concentration is only 1% smaller than its value in solution bulk) increases with the increase in the current density/voltage. At the same time, as the current increases, the thickness of the Nernstian diffusion layer first increases (for $i < i_{lim}$) and then decreases (for $i > i_{lim}$). Figure 6 shows the distribution of interferometric lines. Figure 7 presents the dependence of parameters of different diffusion-layer parts. The points in Fig. 7 show experimental data found from interferograms, the curves were calculated with the use of the two-dimensional “basal” model of transport of ions and water in an ED cell with regard to electroconvection, which was proposed by M. Urtenov et al. [69].

Besides visualization of EC vortices described in [18, 89, 90, 92–96], the diffusion layer structure was studied by means of various indirect methods. The effective EDL thickness was determined as a function of the current density by processing VAC [58, 99], chronopotentiograms [99], and electrochemical impedance spectra [79]. The experimental verification of the data calculated in terms of two-dimensional models (with and without regard to the EC) can be found in [18, 67–69, 94]. In a recent study, S. Nam et al. [100] analyzed the mechanisms of EC induced by either surface conductivity or electroosmotic liquid flow.

MECHANISMS OF MASS TRANSFER INTENSIFICATION

In the electrodialyzer channels at a sufficient distance from the membrane, the forced convection dominates and determines the liquid flow structure even in the overlimiting current modes. At the same time, near the membrane surface, the EC dominates. Mathematical simulation of the ED process makes it possible to distinguish the following main mechanisms of the EC-induced mass-transfer intensification in the ED channel.

(1) Near the IEM surface, a sufficiently complicated system of twin counter-rotating vortices is formed [101, 102] which move like a single whole over the membrane surface under the action of forced convection (Fig. 8). According to the terminology of classical hydrodynamics, such a system is called the vortex cluster [103]. Each vortex pair consists of a large vortex the rotation velocity of which at a sufficient distance from the IEM surface coincides with the forced convection rate and a small vortex which rotates in the

opposite direction. The solution inside the vortex cluster is being mixed so actively that its internal concentration can be assumed to be constant [39, 70, 98]. It deserves mention that as the forced convection rate increases, the small vortex becomes negligible and, thus, all vortices become co-rotating [39, 70].

(2) The system of vortex clusters represents a periodic structure (Figs. 8 and 9b). Bending round them, the forced liquid flow also periodically comes closer to the membrane surface and distances from the latter. Correspondingly, the normal component of forced convection changes its sign along the transverse coordinate. When it is directed from the solution bulk to the membrane, the fresh solution is supplied to the membrane surface, forming regions with the relatively high concentration. On the other hand, when it is directed from membrane to solution bulk, the depleted solution is removed from the membrane surface. Thus, the electroconvective vortex clusters play the important role of barriers to the forced liquid flow by changing its normal component, like the elements of a turbulizing separator or the elements of the surface relief on profiled membranes in the ED apparatus (Fig. 9a).

EFFECT OF THE NATURE OF COUNTERION ON THE EC DEVELOPMENT

This problem was considered in [106, 107]. It was shown that for the fixed ratio i/i_{lim} , the potential drop through the membrane changes in the row $Na^+ > Ca^{2+} > Mg^{2+}$ (Fig. 10). The decrease in the potential drop is explained by the development of EC which intensifies with the increase in the degree of counterion hydration. The higher the degree of ion hydration, the more intense the EC and the more intense the mass transfer for one and the same potential difference. In the row Na^+ , Ca^{2+} , Mg^{2+} , the hydration numbers are 5, 8, 10, respectively [107]. Apparently, this is explained by the fact that the more hydrated ions can carry the larger water volumes, i.e., all other conditions equal, the volume carried along with counterions (electroosmotic transfer) increases with the increase in the hydration number. As the degree of hydration increases, the transition from steady to unsteady EC mode takes place at the lower potential difference, which is expressed as a decrease in the length of the limiting current plateau [106, 107] (Fig. 10) associated with the steady EC mode by the mechanism of EO of the second kind.

In chronopotentiograms, individual potential oscillations were noticed [107] at the times lower than the transition time (τ) as well as periodic oscillations in the quasi-steady state (Fig. 10b). The individual oscillations are observed only for strongly hydrated doubly charged counter-ions Ca^{2+} and Mg^{2+} and correspond to the development of equilibrium EC by the mechanism of EO of the first kind, because the concentra-

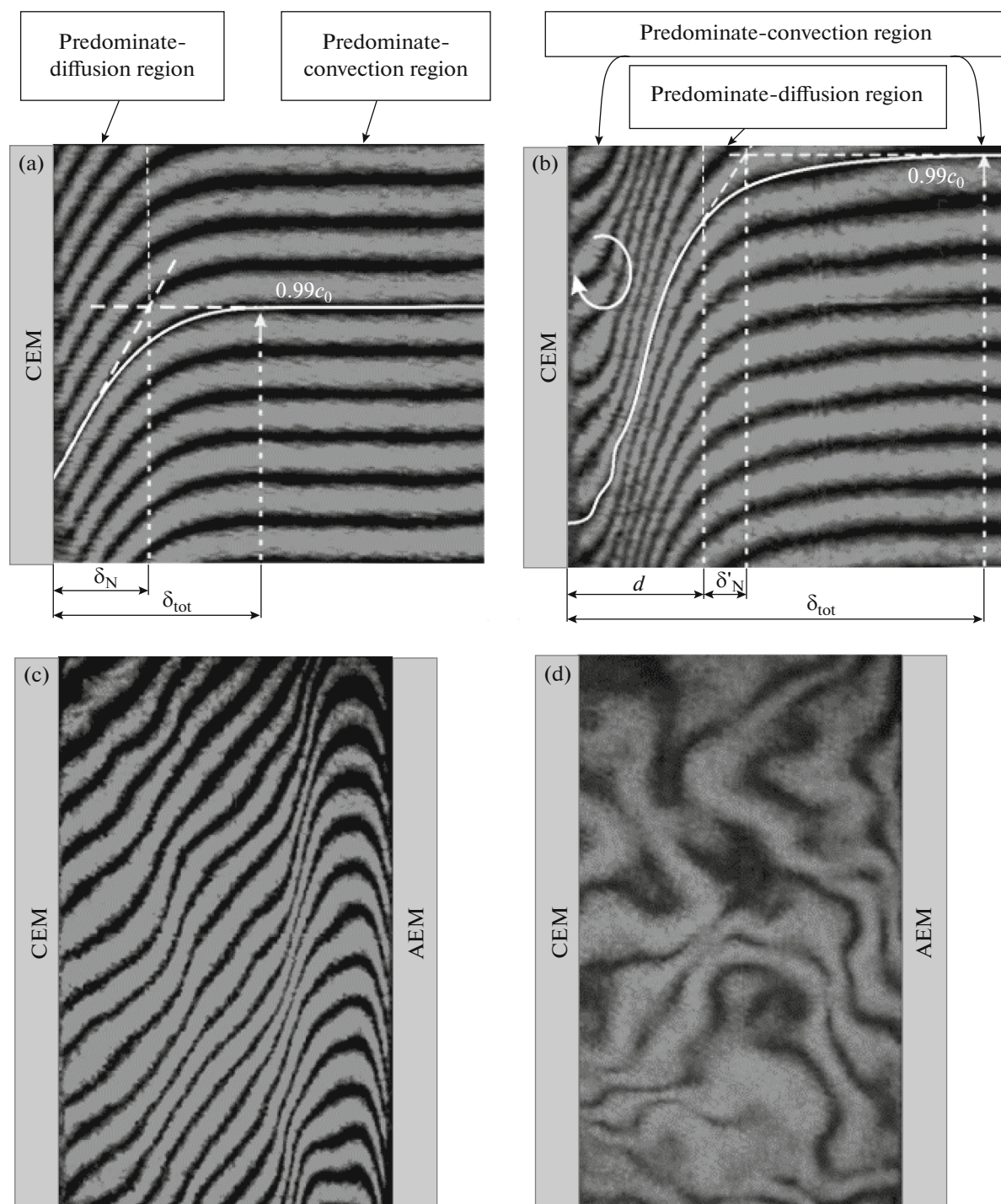


Fig. 6. Experimental interferograms reflecting visually the concentration profiles in a solution flowing between two membranes which form the electrolysizer chamber in the (a) underlimiting mode for the current density $i = 0.5 i_{\text{lim}}$ and in overlimiting modes for (b) $i = 2 i_{\text{lim}}$, (c) $i = 6.5 i_{\text{lim}}$; (a) and (b) are the cases where the channel is formed by two CEM (MK-40); in (c) and (d) both CEM (MK-40) and AEM (MA-40) were used. The evaluation of the following values is illustrated: the total diffusion layer (DS) thickness δ_{tot} ; the effective Nernstian local DS thickness δ_N for $i < i_{\text{lim}}$; the thicknesses of the region with predominant diffusion mechanism of ion delivery δ' and the region of electroconvective instability d for $i > i_{\text{lim}}$. (d) The interaction of EC vortices formed in the depleted layer near CEM and AEM leads to chaos. Adapted from [98].

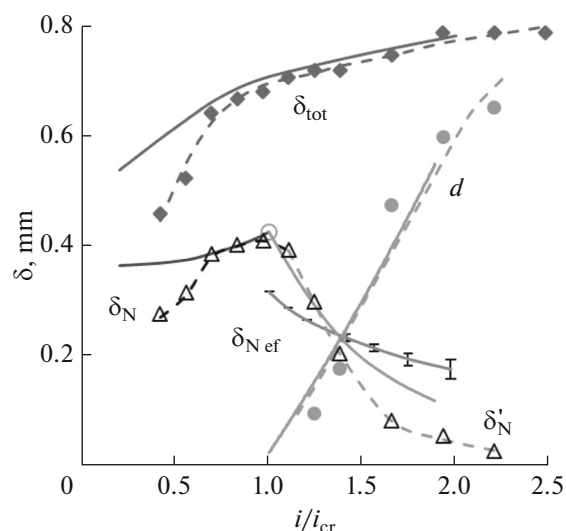


Fig. 7. Dependence of the thickness of different diffusion-layer regions on the current density normalized to its critical value i_{cr} corresponding to the appearance of unsteady EC by the Rubinstein–Zaltzman mechanism [28]. These results correspond to the section at a distance $y = 2.7$ cm from the inlet to the ED channel. Points are the experimental data obtained by processing the concentration profiles when laser interferometry was used; dashed line are the trend lines [97]. Solid curves are calculated in terms of the “basal” 2D model [69]. $\delta_{N\text{ ef}}$ is the effective thickness of DS. Adapted from [98].

tion at the membrane surface in this moment is still relatively high so that no extended SCR is formed. The oscillations in the quasi-steady mode are associated with the Rubinstein–Zaltzman instability of nonequilibrium EC and their period and amplitude increase in the row: $\text{Na}^+ < \text{Ca}^{2+} < \text{Mg}^{2+}$, i.e., with the increase in the degree of hydration of counterions.

THE ROLE OF MEMBRANE'S SURFACE PROPERTIES

The surface properties substantially affect the parameters of electroconvection. To gain insight into the nature of this effect, we formulate the following conditions:

—the realization of intense electroconvection requires the presence of electric field with two components: the normal component which creates the extended space charge region and the tangential component which is responsible for the SCR sliding over the surface [63, 64, 108, 109]. Note that even for the homogeneous surface when EC is developed by the mechanism of electrokinetic instability of Rubinstein–Zaltzman [28], the tangential component is present because the current lines bend round the regions of depleted solution at the membrane surface (Fig. 9b);

—the tangential component appears on the boundaries of surface regions with the high and the low con-

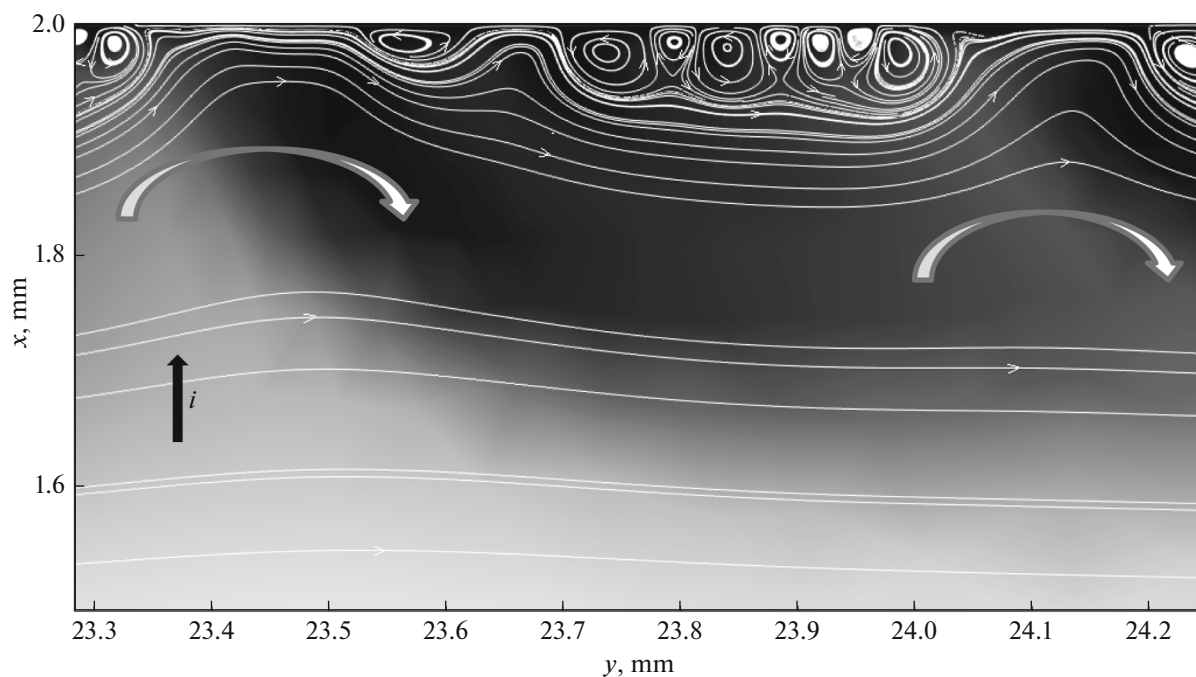


Fig. 8. Concentration field (concentration value is shown by shade of grey) and the lines of liquid flow (white lines), calculated in terms of the 2D basal model [69] based on the system of equations of Nernst–Planck–Poisson–Navier–Stokes (NPP–NS). Curved arrows show the direction of convective transfer of ions. Adapted from [98].

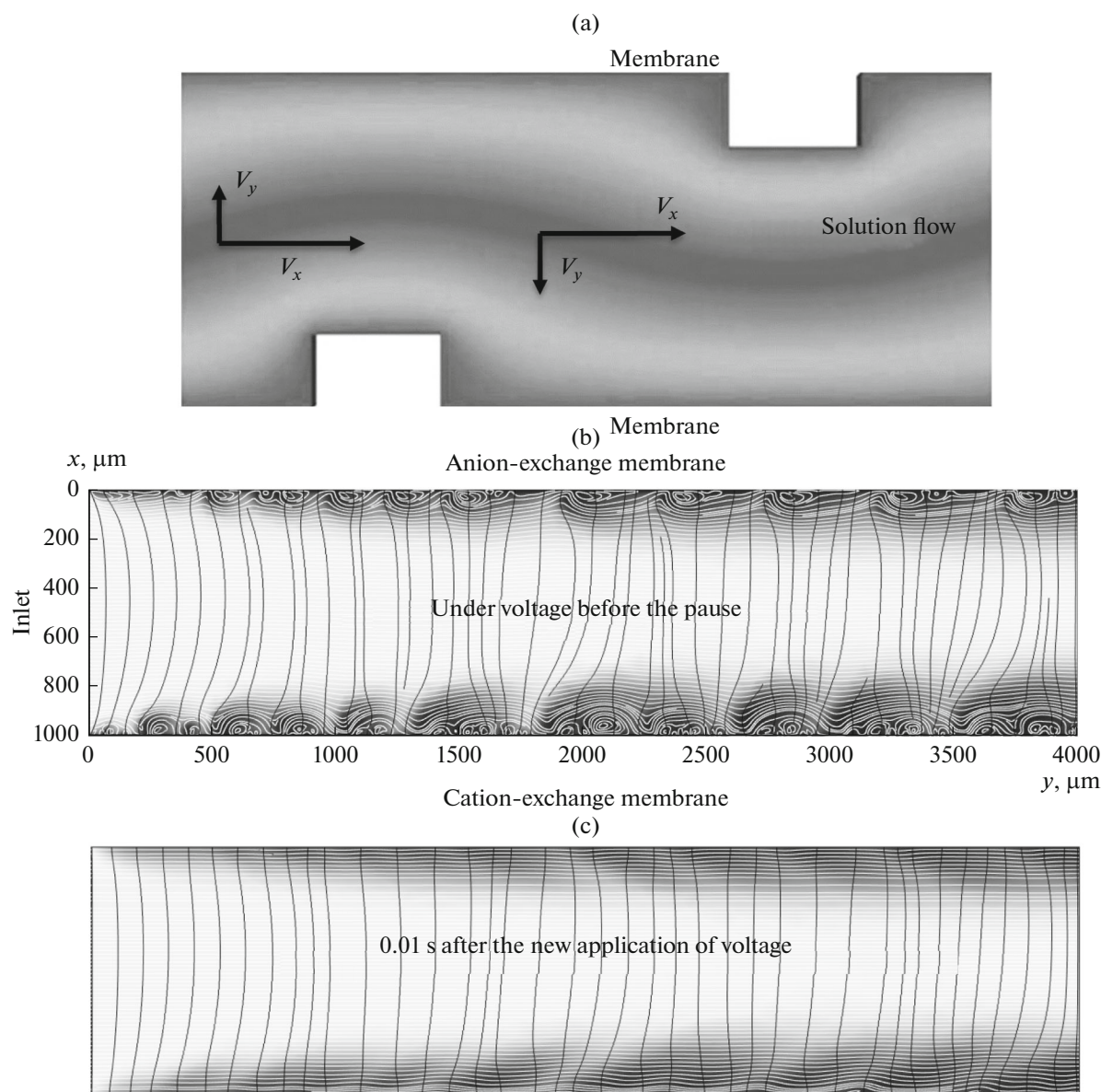


Fig. 9. (a) Distribution of liquid flow velocities (shades of grey) in an elementary section of the channel of an electrodyalyzer with profiled membranes; adapted from [104]. (b, c) Distribution of solution flow lines (white lines), electric current lines (black lines), and concentration (shades of grey) in the desalination channel formed by anion and cation-exchange membranes. (b) shows the quasi-stationary state at the potential drop of 2 V per pair of chambers; (c) shows the state in 0.01 s after the new switching of voltage of 2 V which follows the pause of 0.5 s. Adapted from [105].

ductivity and also at surface ridges and valleys (Fig. 11). Thus, the distance between such boundaries along the membrane surface essentially determines the distance between individual vortices;

—the sliding of EC vortices over the surface becomes easier if the slide surface is relatively hydrophobic;

—insofar as EC vortices tend to acquire the circular form (in their two-dimensional section), then to obtain vortices capable of substantially affecting the mass transfer (i.e., having the size comparable with the diffusion layer thickness δ), the size of heterogeneities

on the membrane surface should also be comparable with δ [28, 36, 45, 63, 109, 110].

Below we show which effects can be attained by varying electric heterogeneity, the degree of hydrophobicity, or by forming a relief on the membrane surface.

(1) The presence of electric heterogeneity on the IEM surface gives rise to the so-called “funnel effect” [63, 111], which consists in concentration of current lines in solution near the conducting surface regions (Fig. 11a). This effect was taken into account when finding the theoretical value of the limiting current

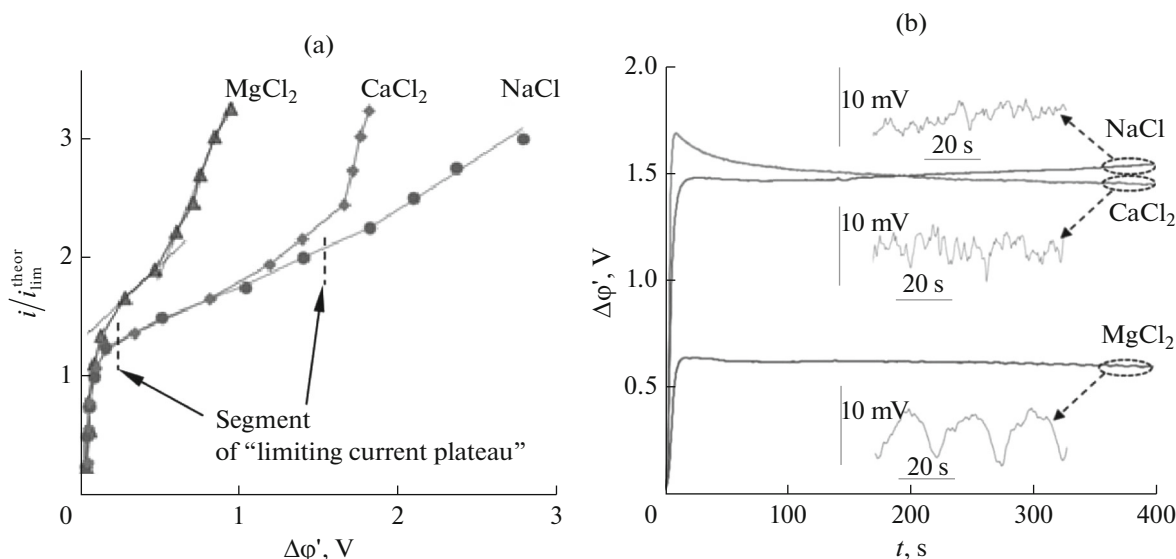


Fig. 10. VAC of membrane MK-40MOD in 0.02 M solutions of NaCl, $CaCl_2$, and $MgCl_2$. Points correspond to potential drop values (corrected for the ohmic component) in the quasi-stationary state of the system at a fixed current density. The VAC segment called in the literature the "limiting current plateau" [28] is shown for the case of NaCl. Adapted from [107].

density [63, 112, 113] and also of the transition time [112, 114] by solving the Nernst–Planck equations under the conditions of local electroneutrality of solution. It was shown that in contrast to membranes with the homogeneous surface, the concentration of current lines at the conducting regions of an electrically heterogeneous membrane leads to the local decrease in concentration, which reduces the limiting current density and the transition time and also increases the resistance of the system.

At the same time, the funnel effect determines the appearance of the tangential current component, which favors the EC development by the electroosmotic mechanism [45, 108, 115]. It was shown theoretically [92, 116] that the presence of nonconducting regions on the IEM surface makes it possible to intensify the EC and, as a result, increase the overlimiting current density as compared with homogeneous membranes, although a part of the heterogeneous membrane surface remains nonconducting (Fig. 12).

(2) The hydrophilic membrane surface prevents the development of electroconvective vortices because of the adhesion of water molecules which reduces the liquid velocity in the tangential direction. The decrease in hydrophilicity/increase in hydrophobicity of the surface gives rise to the effect of solution "slipping" over this surface (Fig. 11c), which in turn intensifies the EC. As proved experimentally, this can be achieved by covering the IEM surface with an ion-exchange material based on a hydrophobic matrix, e.g., a Nafion [117, 118] or MF-4SK film [107] (the effect can be intensified by adding carbon nanotubes into the coating [119] (Figs. 13a and 13b)) and also by

the spot deposition of some uncharged inert hydrophobic material [120] (Figs. 13c and 13d).

(3) As was mentioned above, the development of a wavy surface (profiled relief) favors the increase in electroconvection. According to estimates [28], 10% surface curvature should lead to 30% increase in mass transfer. In this case, the threshold voltage at which the unsteady EC starts should be lower. In the VAC, this should be manifested as a decrease in the length of the plateau (inclined) which corresponds to the current density equal or somewhat exceeding the limiting value when the EC proceeds in the steady mode.

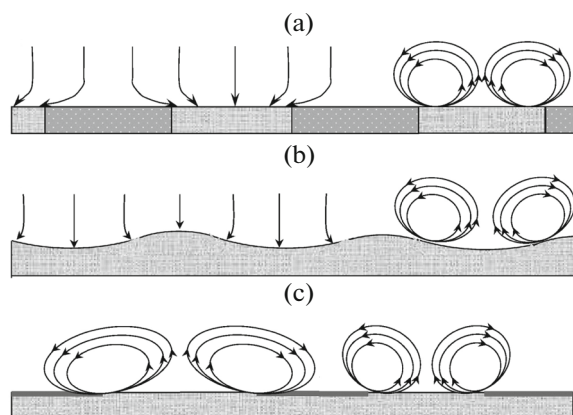


Fig. 11. Illustration of the distribution of current lines (arrows) and electroconvective flows near the IEM with the (a) electrically and (b) geometrically nonuniform surface and also (c) in the presence of regions with different degree of hydrophobicity. Adapted from [108].

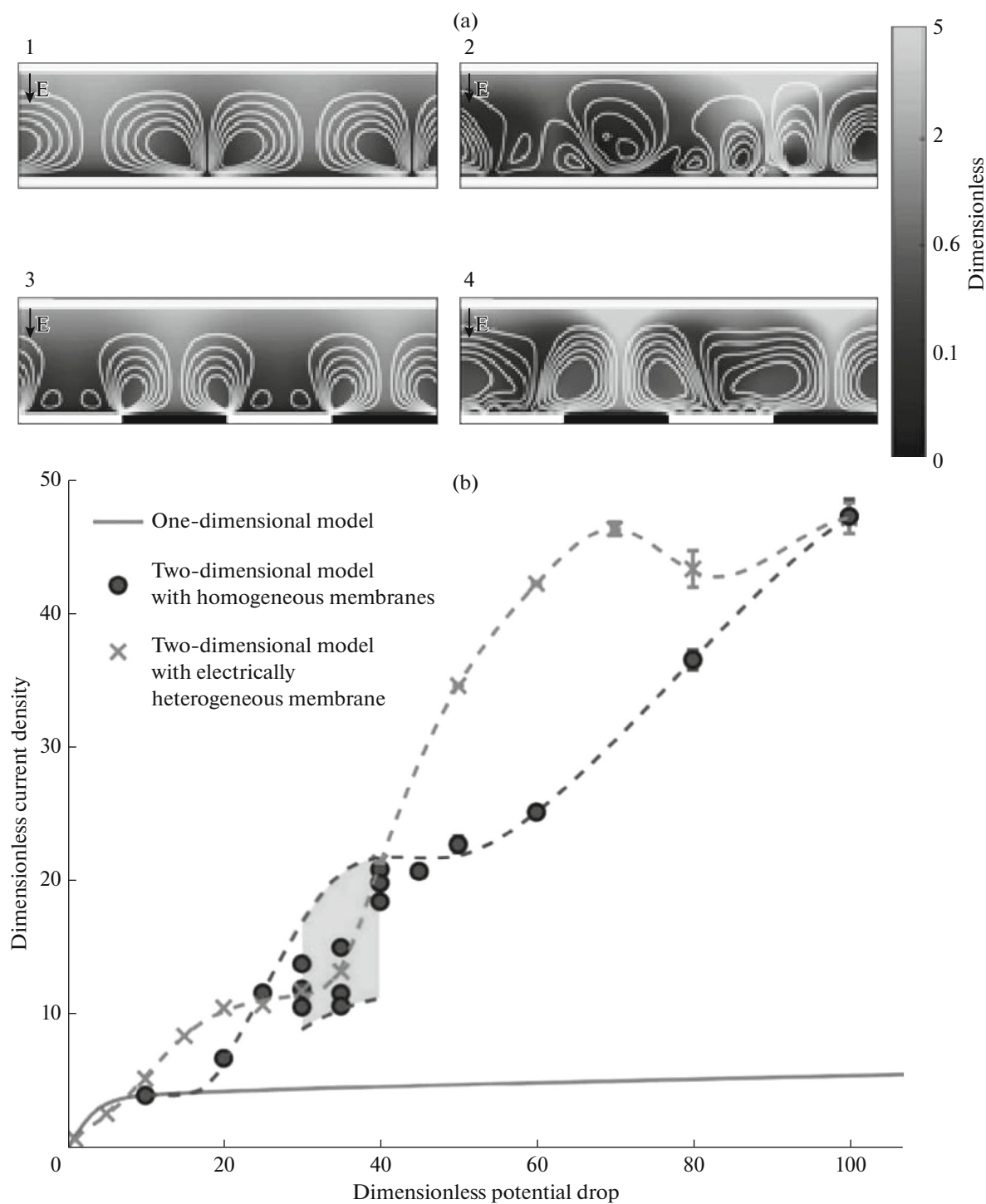


Fig. 12. The effect of electrical heterogeneity of the membrane surface (a) on the development of EC for the potential drop equal to (a1 and a3) 0.5 V and (a2 and a4) 1.5 V and also (b) on the form of VAC. Theoretical VAC correspond to two models based on equations NPP–NS: one-dimensional model with disregard of convective flows (1D) and two-dimensional model with regard to convection (2D) with homogeneous membranes (cases a1 and a2) and an electrically heterogeneous membrane (cases a3 and a4). Adapted from [116].

For the first time, the ED apparatus with profiled membranes was developed by N. Gnusin et al. [122]. The profile represented alternating parallel strips

(ridges) where the membrane surface served as the base. A. Belobaba et al. [123] showed that the use of such type of profiled membranes provides a consider-

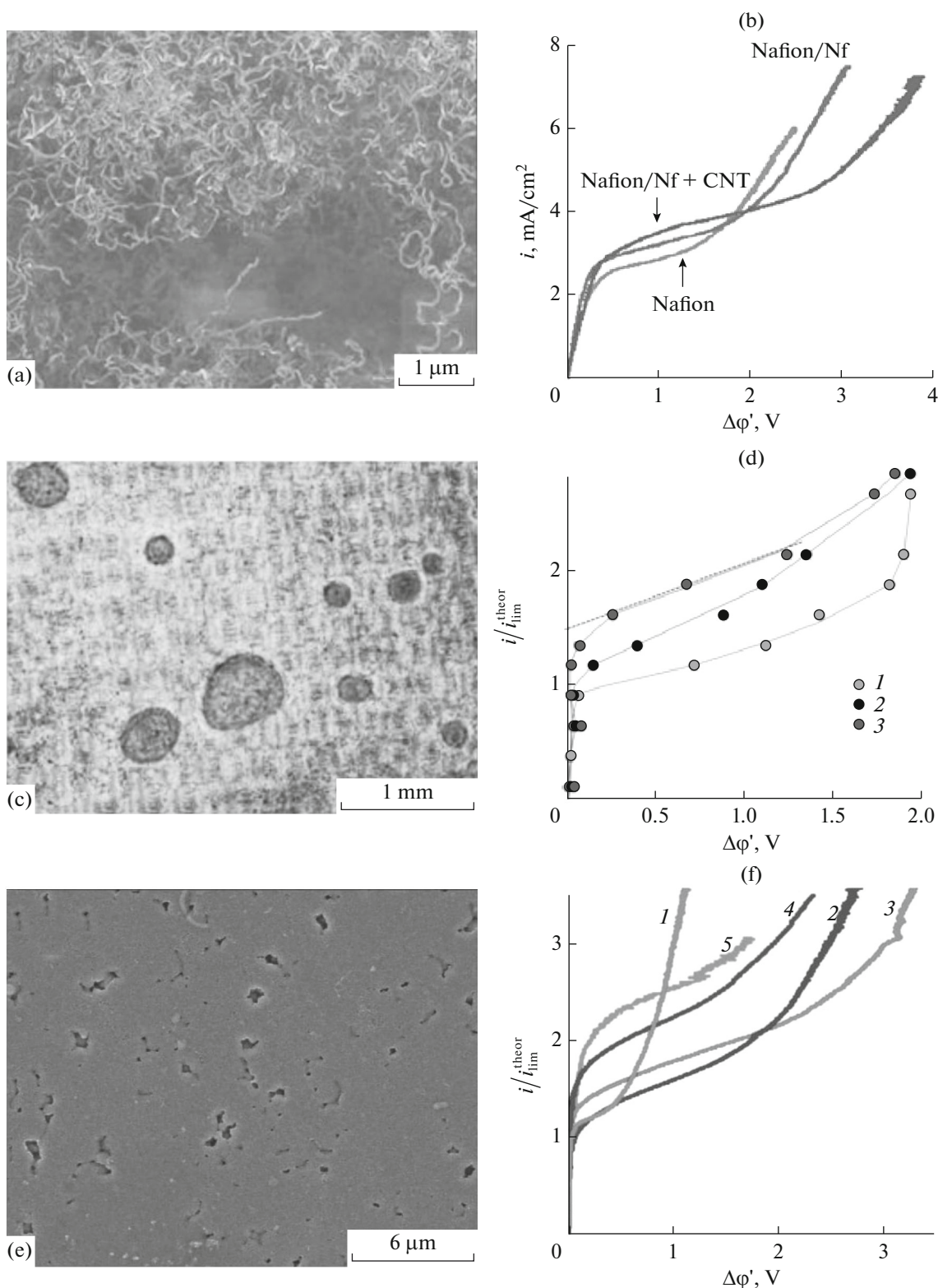


Fig. 13. Microimages obtained on a scanning electron microscope; (a) Nafion membrane covered with a thin Nafion layer containing nanotubes; (c) anion-exchange AMX membrane with hydrophobic spots deposited on its surface; and (e) cation-exchange CMX membrane with caverns formed as a result of overlimiting current passed for 100 h. On the right, the VAC of the corresponding membranes are shown: (b) VAC for unmodified Nafion membrane and Nafion membranes covered by a Nafion layer and also by a Nafion layer containing nanotubes, adapted from [119]; (d) VAC of unmodified membrane AMX (no. 1) and its two modifications obtained by coating 8% (no. 2) or 12% (no. 3) of its surface with hydrophobic spots. The potential drop values in VAC correspond to stationary potential drops in chronopotentiograms. Adapted from [120]; (f) VAC of the (1) initial membrane CMX and (2–5) its samples after 10, 25, 100, and 150 h treatment under current, respectively. Adapted from [121].

Table 1. Surface properties of anion-exchange membranes [136]

Sample	I AMX-SB _{in}	III AMX-SB _{mod1}	IV AMX-SB _{mod2}	II AMX-SB _{cor}
Flow potential, mV/bar	15.0	4.1	−0.4	−0.8
Zeta-potential, mV	248.0	67.0	−6.1	−14.0
Surface charge, $\mu\text{C}/\text{cm}^2$	112.0	2.9	−0.2	−0.5

able gain in the mass transfer velocity in the overlimiting current mode. Later, the membranes with analogous profile (formed as strips with rectangular, triangular, or trapezoid cross-section) were proposed and studied in several publications [124–126].

J. Balster et al. [80] studied the VAC of several membranes with smooth or modified surfaces. On the surface of the latter, a relief was formed which represented regular alternating ridges and valleys with their length substantially exceeding their width (this relief can be represented as a system of parallel rectangular ridges lying on the smooth surface). The ridge height was from 10 to 60 μm , their width was about 50 μm . The width of valleys varied from 50 to 550 μm . It was found that the surface relief exerted a substantial effect on the parameters of overlimiting transport, particularly, on the length of the limiting current plateau, which decreased with the increase in the ridge height. The reduction of the plateau length reached 60%. The limiting current density (determined from intersection of tangents to the initial and plateau regions) was observed to increase by about 30%.

Profiling of the IEM surface by pressing in order to form individual conic ridges with the height of about 0.6 mm [127] made it possible to increase the mass transfer rate in the overlimiting state (at the voltage of 2.5 V per chamber pair) approximately 4 times as compared with the case of smooth membranes and a separator in the form of a wire net with nonconducting wires [38, 128]. The sufficient advantage from the use of profiled membranes is provided simultaneously by several synergistic effects [38, 126, 129]: the turbulization of the forced liquid flow in channels of the ED apparatus without using spacers, the development of the conducting membrane surface, the appearance of the tangential electric force intensifying the EC.

It deserves mention that the surface of series-produced nonprofiled membranes is not ideally smooth [130–132], which considerably affects their properties. As was shown theoretically [65, 108, 115], the development of the IEM roughness favors the appearance of the tangential component of current and the development of EC by the mechanism of EO of the second kind. By 2D simulations with the use of a system of conjugated equations NPP–NS, S. Pham et al. [65] showed that the formation of a wavy structure with the

ridge height one-thousand-times-smaller than the wavelength substantially lowered the threshold of the appearance of unsteady EC by the mechanism of Rubinstein–Zaltzman. It was shown experimentally [121] that the reason for the substantial increase in the overlimiting current could be the formation of micrometer caverns (appearing as a result of electrochemical corrosion of the polymer) on the relatively smooth membrane surface [121] (Figs. 13e and 13f). The increase in the fraction of nonconducting surface from 0.18 to 0.25 and the increase in the difference between the highest and the lowest points on the surface from 5 to 15 μm led to the substantial (~ 1.5 -fold) increase in the current density and a considerable decrease in the rate of generation of H^+ and OH^- ions [133]. Note that the decrease in the rate of generation of ions H^+ and OH^- on the interface also favors intensification of electroconvection [63, 97, 133–135].

(4) As was shown above, hydrophobization of the membrane surface allows the EC to be intensified at relatively high potential differences. At the same time, it is known that the increase in the surface charge (resulting in the higher hydrophilicity) intensifies the classical EO of the first kind [136]. In [136], this apparent contradiction was resolved. An anion-exchange membrane Neosepta AMX-SB (with the high surface roughness [130]) and its three modification differing by the surface charge were studied (Table 1). By means of chronopotentiometry and voltammetry, it was found that for small potential differences (the normalized potential difference $\Delta\phi'$ in which the ohmic component is ignored was lower than

100 mV) and the current densities of up to $0.8i_{\text{lim}}^{\text{theor}}$, the intensity of EC was the higher for the higher surface charges (Fig. 14a). For the relatively larger potential differences ($\Delta\phi' > 0.5 \text{ V}$) and current densities ($> i_{\text{lim}}^{\text{theor}}$), the EC developed more actively near membranes with relatively more hydrophobic surface (Fig. 14a). Such a behavior of membranes is explained by the different mechanisms of EC taking place at small and large values of potential difference/current density. In the former case, the EC mechanism is the EO of the first kind, where the quasi-equilibrium EDL with the charge determined by the surface charge shifts under the effect of the external force. In the lat-

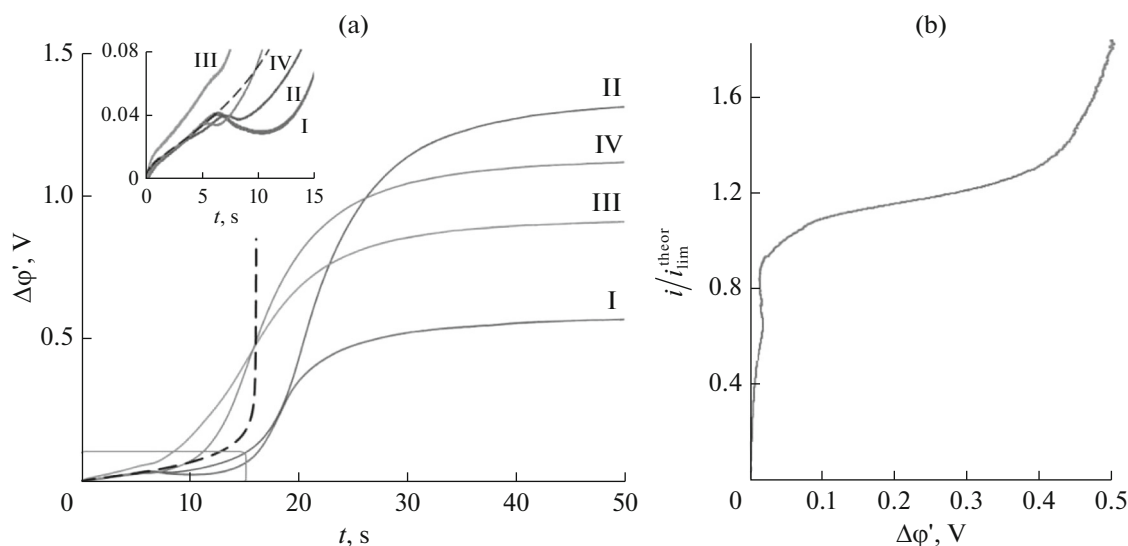


Fig. 14. (a) Initial segments of experimental (lines) chronopotentiograms for anion-exchange membrane AMX-SB and its three modifications (numbers of curves correspond to their numbers in Table 1) for the current density $i = 4.5 \text{ mA cm}^{-2} (=1.44i_{\text{lim}}^{\text{theor}})$; theoretical curve (dashed line) calculated for "ideal" membrane in terms of 1D three-layer model with disregard of EC (adapted from [136]); (b) VAC of membrane AMX-SB.

ter case, as we pass to the EO of the second kind, the chief role in EC is played by the extended SCR with the charge independent of the surface charge. The solution slipping near hydrophobic regions becomes the dominating factor.

The specific form of chronopotentiograms for the AMX-SB (I) membrane and its modifications II, III, and IV observed at the small times/potential differences deserves mention. The formation of electroconvective vortices gives rise to a decrease in the system resistance manifested as a decrease in the potential difference ($\Delta\phi'$) and the appearance of a local maximum (Fig. 14a). The depth of the $\Delta\phi'$ decrease is the maximum for the AMX-SB membrane with the maximum surface charge. For membrane III with the surface charge minimum in the absolute magnitude, $\Delta\phi'$ increases with time even faster than in the theoretical curve calculated for the "ideal" homogeneous membrane (1D model) with disregard of EC. Such behavior is apparently caused by the electric heterogeneity of the surface due to the nonuniform distribution of the modifier over the surface (funnel effect). It is interesting that the "anomalous" decrease in the potential for AMX-SB can be also observed in the quasi-equilibrium voltammetric curve in which the current is scanned slowly in time and the potential drop is recorded (Fig. 14b): in the segment from 0.6 to $0.8i_{\text{lim}}^{\text{theor}}$, as the current increases, the potential drop not only stops to increase but even decreases. This peculiarity points to the presence of considerable EC by the mechanism of EO of the first kind, which was never discussed in the literature. The fact that for IEM with certain surface properties, one can reach rela-

tively high current densities (close to the theoretical limiting value) at a very small normalized potential drop (about 100 mV), allows one to expect that this effect can be used in practice.

EFFECT OF PULSE CURRENT MODE ON THE DEVELOPMENT OF ELECTROCONVECTION AND MASS-TRANSFER RATE

Pulse current was effectively used for a long time in electroplating for improving the physicochemical and functional properties of electroplates (to increase the coating adhesion with the support, decrease its porosity, increase the hardness and wear resistance, increase the corrosion resistance, etc.) [137–140]. In the practice of membrane processes, the use of such current modes was initiated relatively recently [141]. The main effects in this case are the increase in the mass transfer rate [142, 143] and, which probably has the practical importance for ED, the inhibition of membrane fouling [141, 144–146]. Moreover, in the pulse mode, the rate of water dissociation at the membrane/depleted solution interface decreases [5, 6, 147, 148].

When pulse current is used, the mass transfer rate increases with the increase in the pulse current frequency. This fact is well known in electroplating where the pulse current with frequency of up to 500 Hz is used [149]. For the higher frequencies, the parasite process of EDL charge–discharge becomes noticeable. In an ED membrane cell [143] which represents a set of alternating cation and anion-exchange membranes which form the chambers of desalination and

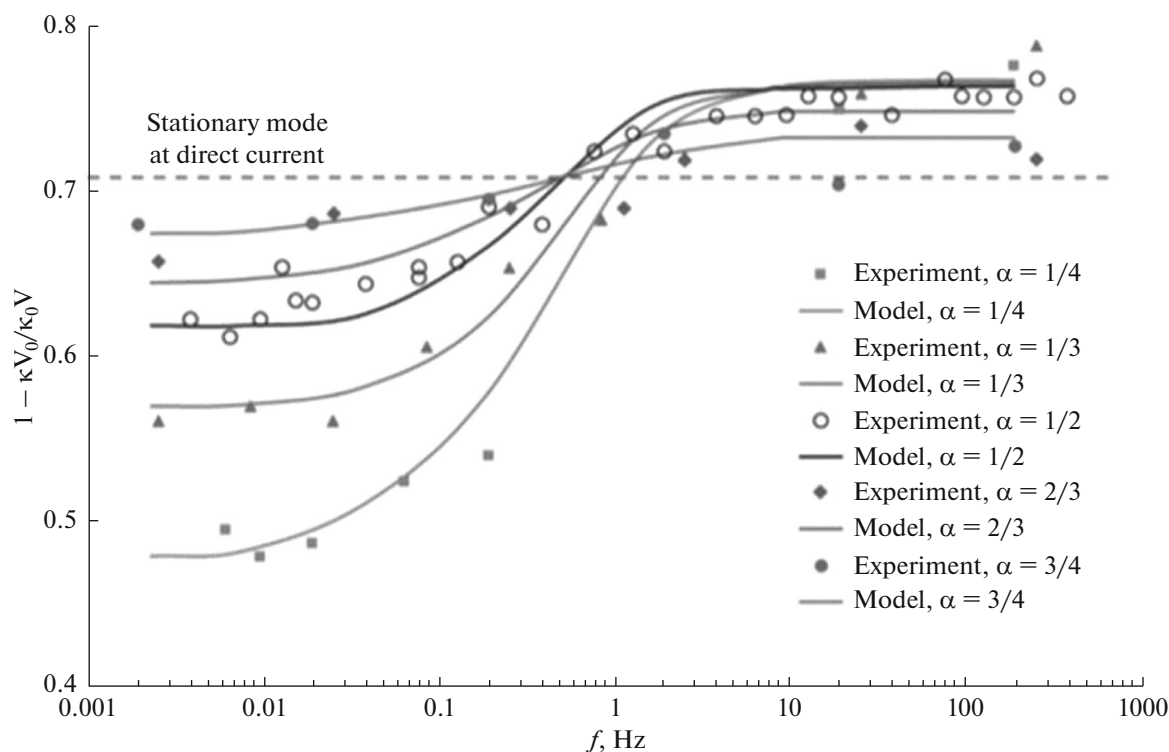


Fig. 15. Dependence of the degree of desalination on the pulse current frequency at different duty factors α . The experiment was carried out at underlimiting current densities under the condition that the cycle-averaged potential drop remains constant (equal to 0.6 V per chamber pair). Adapted from [143].

concentration, the charge-discharge currents become noticeable at frequencies >150 Hz during desalination of sodium chloride solutions with initial concentration of 0.1 M [143]. Figure 15 shows the dependence of the mass-transfer rate expressed as the degree of desalination of a fixed volume of sodium chloride solution with the initial concentration of 0.1 M in the circulation mode for 10 min. The experimental data are compared with the results of calculations in terms of the 1D model based on the Nernst–Planck equations and the condition of electroneutrality. In contrast to conventional approaches used for describing the pulse current modes in electroplating [137, 150, 151] and also in membrane systems [142, 152, 153], the authors of [143] modeled the transport not only in the depleted diffusion layer but also in the enriched diffusion layer and in the membrane. This allowed the experimentally known potential difference per chamber pair to be calculated.

According to Fig. 15, the experiment and calculations showed that the mass transfer rate stops to increase at frequencies >100 Hz. As the frequency decreases, an asymptotic limit is observed (reached at approximately 0.005 Hz) in which the mass-transfer rate reaches its minimum. The presence of the upper and the lower limits is explained by the fact that the length of diffusion relaxation of concentration profile in solution varies. As was shown earlier by N. Ibl [137],

in the pulse current mode, the diffusion layer can be divided into two parts. In one part remote from the interface, the concentration profile is stationary, whereas in the other part (at the surface of electrode or membrane), it oscillates. During the pause, the partial relaxation of the concentration profile at the membrane surface occurs and the solution resistance substantially decreases. In the beginning of the next cycle, the relatively high voltage is applied to the system with the low resistance, as a result of which a high current density is reached. In average, during one cycle, the gain in the mass-transfer rate (at the invariant time-averaged voltage) is associated with the high contribution of electromigration transport at the second switching of voltage. The upper limit of mass-transfer rate is reached when the thickness of the oscillating part of the diffusion layer becomes negligible as compared with its total thickness. The lower limit is reached when the length of the region in which the diffusion relaxation occurs exceeds the diffusion layer thickness [143].

One peculiarity of the pulse mode is that the high current density is reached at the relatively low concentration polarization of the membrane system. At sufficiently short cycles, the concentration of salt ions at the interface has no time to reach sufficiently low values at which the intense water dissociation can occur. For this reason, the rate of generation of H^+ and OH^-

ions on the IEM/solution interface is substantially lower as compared with the stationary mode at the same average current density [147]. A consequence of the lower rate of generation of ions H^+ and OH^- is the decrease in the sedimentation rate (or its total absence) in desalination chambers [137–140]. Such type of sedimentation is caused by the interaction of certain cations (Mg^{2+} , Ca^{2+}) with OH^- ions formed as a result of water splitting at the surface of a cation-exchange membrane [5]. Hydroxides of these ions have very low constant of heterogeneous equilibrium in saturated solutions, known as the solubility constant. For $Mg(OH)_2$ at $t = 25^\circ C$, this constant is equal to 1.8×10^{-11} . For the local pH 10, the sediment can appear even at the magnesium-ion concentration of about 10^{-3} M.

Experimental studies [5, 147] showed that the pulse current mode is still more efficient for the overlimiting current densities. In this case, one can raise substantially the mass transfer rate, avoiding, as before, the undesirable water splitting and sedimentation. At the same time, when passing to the overlimiting current densities, the optimal frequency of pulse current shifts, i.e., becomes much lower, of the order of magnitude of 1 Hz in place of 10–50 Hz for underlimiting currents. As was shown by simulations based on conjugated equations of NPP–NS [105], the EC plays the important role at $i > i_{lim}$. Whereas for underlimiting currents the gain is associated with short-term delay in the growth of system's resistance as compared with the increase in the current taking place after the pulse is applied, for overlimiting currents the nonuniform concentration field initiates the fast development of EC, making easier the mass exchange between the membrane surface and the solution bulk. According to calculations [105], after 0.5 s pause (during which the EC vortices fade out but the nonuniform concentration field is leveled incompletely, Figs. 9b and 9c), the new EC vortices are generated as soon as in several hundredths of s.

During the development of EC, the low solution resistance attained during the pause keeps longer after the new pulse is applied, because the EC vortices that appear almost immediately (Fig. 9c) mix the solution near the surface. Nonetheless, the vortex formation requires a certain time which results in the fact that the optimal pulse length for overlimiting currents turns out to be higher as compared with underlimiting currents.

It is interesting that back in 2001 N. Mishchuk et al. [153] hypothesized that the gain in mass transfer for overlimiting pulse modes can be explained by the fact that during the pause the electroconvective vortices do not fade completely but continue to rotate due to inertia forces. Calculations [105] showed that this prediction is true with certain reservations. Indeed, the vortices continue to rotate during the pause. However, the vortex lifetime during a pause is several tenth fractions

of second, which is substantially lower than the pause length during which the current mode is close to optimal. Moreover, according to the authors of [153], the reason for the persistence of the vortex life after the current/voltage has been switched off is not the inertia forces (these forces persist only for one-two hundredths of second [105]) but the fact that after switching off the voltage, the electroosmotic liquid fluxes hold out in the membrane system for a certain time. These fluxes are associated with the nonuniform distribution of counterion concentration along the transverse coordinate inside the EDL (Figs. 9b and 9c). In the regions where the solution is strongly depleted and the EDL thickness is large, the space charge at a certain distance from the membrane is also high, whereas in the region with the small EDL thickness, it is low. The transverse diffusion of counterions inside the SCR gives rise to electric current, which in turn induces electroosmotic sliding of the liquid. Thus, the nonuniform distribution of the volume electric force appears, which feeds the vortices. Because the relaxation time of the concentration field is sufficiently long (several seconds), the feeding of vortices with the chemical energy of this nonuniform field can continue for a sufficiently long time. Note that the analogous principle of conversion of chemical energy to electric energy underlies the operation of reverse ED devices for power production [154] in places where a river inflow into the sea, which use the difference in the salt concentration in the river and in the sea.

CONCLUSIONS

Electroconvection is a phenomenon which can be successfully used in modernizing both the processes of separation of dissolved substances and the processes of solvent transport. In separation processes with the use of membranes, the issue of interest is the possibility of intensifying the rate of ion exchange between solution bulk and the membrane surface. Moreover, EC can efficiently suppress the sedimentation of mineral substances and the formation of organic/colloid deposits on the membrane surface and in its pores. Both effects were proved experimentally, the mass transfer under EC conditions is described theoretically. The modern studies are aimed at elucidating the origin of the relationship between the surface properties and EC intensity and mass transfer rate. Different electric modes are considered: relatively small and large potential differences/current densities, pulse current modes. Many interesting effects revealed in the pulse mode (the gain in mass transfer rate, the decrease in the rate of current-induced chemical reactions at the interface, suppression of fouling), even if are already sufficiently substantiated, still require more detailed theoretical explanation.

The fast progress in understanding the origin of electroconvection and its role in membrane processes of separation and also in micro-nanofluid systems

deserves mention. Twenty years ago the mere existence of the electroconvection effect in membrane systems was called into question, ten years ago scientists understood that by changing certain properties of the membrane surface one can increase the mass transfer rate due to electroconvection. At present, the main vector of studies is aimed at optimizing both the surface properties and the electric mode parameters by using electroconvection, which would allow one or another advantages to be achieved in organization of transport processes.

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